Jeffrey Walker

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Summary

Robotics and embedded systems engineer with hands-on experience across cable-driven soft robotics, embedded medical devices, industrial automation prototyping, computer vision, and simulation-backed control work. Strongest evidence comes from real hardware integration, microcontroller-based actuation, sensor and motor interfacing, and building systems that connect mechanical design, firmware, and experimental validation.

Skills

Embedded & Robotics: Arduino, ESP32, Raspberry Pi, NVIDIA Orin Nano, actuator and sensor integration, motor control, serial interfaces, PLC logic, electromechanical debugging

Software & Tools: Python, C/C++, MATLAB, Git, OpenCV, ROS, MuJoCo, Gazebo, Docker

Mechanical & Build: SolidWorks, GD&T, 3D printing, CNC, lathe, drilling, prototyping, machine design, mechatronic integration

Education

Drexel University M.S. Mechanical Engineering, Robotics, Systems & Controls focus 2023–2025
Thesis: *Advanced Flight Control for Nonlinear Aircraft Models (NASA F-18)* Philadelphia, PA

Kwame Nkrumah University of Science and Technology B.S. Automotive Engineering 2019–2023
Project: *Volkswagen Transporter redesign and electrification* Kumasi, Ghana

Experience

Controls and Embedded Systems Intern – Medical Devices *T.I. Inc, Pennsylvania* Sep 2024 – Feb 2025

- Developed embedded control and interface logic for neonatal medical-device subsystems using Arduino, ESP32, C/C++, and Python.
- Integrated sensors, actuators, displays, and motor drivers into electromechanical prototypes with real-time response constraints.
- Implemented watchdog behavior, fail-safe logic, and structured test workflows to improve prototype reliability in a regulated environment.

Industrial Automation Systems Researcher *Drexel University – Prof. Dela Fuentes, Philadelphia, PA* May 2024 – Present

- Conceptualized closed-loop industrial automation architectures using sensing, actuation, and PLC-oriented logic.
- Built prototype hardware using additive and subtractive manufacturing workflows at the Drexel Innovation Studio.
- Supported system design at the intersection of controls, automation, prototyping, and manufacturability.

Automotive Engineering Workshop Staff / Engineer *KNUST, Kumasi, Ghana* Nov 2022 – Sep 2023

- Supported workshop engineering activities spanning manufacturing, maintenance, prototyping, and vehicle-system troubleshooting.
- Used CNC, lathe, and drilling machines to manufacture or refine automotive components, contributing to a reported **10% reduction in part failure rates**.
- Supported maintenance and diagnostic work on more than **60 vehicles**, strengthening hands-on mechanical and systems intuition.

Selected Projects

SpiRob Cable-Driven Soft Robot *Embedded Control / Soft Robotics / Mechatronics* 2024

- Developed a spinal-inspired soft robot with cable actuation, stepper-motor control, and modular embedded logic for wave-like motion experiments.
- Combined physical prototyping, embedded actuation, and MuJoCo-oriented modeling to connect hardware work with future autonomy and control development.

Vision Safety Monitoring System *Computer Vision / OpenCV / Monitoring* 2025 – 2026

- Built an OpenCV-based monitoring pipeline with interactive no-go zones, posture labeling, event detection, and alert logging.
- Designed the workflow for live or file-based execution with practical operator interaction and deployment-oriented structure.

Stereo SLAM Pipeline for UAV Workflows *SLAM / Perception / Evaluation* 2025

- Built a stereo SLAM workflow with KITTI-style data recording, pluggable backend structure, and trajectory evaluation.
- Structured the project around pose visualization, alignment, and interpretation for robotics-navigation experimentation.

Nonlinear F-18 Flight Control *Controls / Simulation / Evaluation* 2024 – 2025

- Built validation workflows around a nonlinear aircraft model using MATLAB/Simulink, FlightGear, and controller comparison.
- Reinforced systems-level thinking across modeling, software, instrumentation, and interpretation.

Honors

- **Dean's Fellowship**, Drexel University 2023 – 2025
- **Leroy L. Resser Endowment Scholarship** 2024
- **College of Engineering Student Ambassador**, Drexel University 2024 – 2025